5633032

B.Tech. DEGREE EXAMINATION, SEPTEMBER 2021.

Third Semester

Computer Science Engineering

DATA STRUCTURES

Time : Three hours | Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. How data structures are categorized?

2. Explain time complexity.

3. How De queues are different from queues.

4. Convert the following infix expression to prefix and postfix expressions (A+B)*(C–D).

5. List the applications of B tree.

6. How many leaf nodes will be there for a binary tree of n nodes?

7. List the applications of graphs.

8. Explain the representation of the graph.

9. Define symbol table.

10. Explain external sorting.

SECTION B — (5 × 11 = 55 marks)

Answer ALL the questions.

All questions carry equal marks.

UNIT I

11. Write the bubble sort technique. Illustrate the given sort using simple example.

Or

12. Discuss in detail about algorithmic notation.

UNIT II

13. Explain various operations of stack.

Or

14. Elaborate on memory representation of linked list. Also explain the types of linked list.

UNIT III

15. Discuss the various types of trees. Also list th basic terminologies of binary tree.

Or

16. Discuss in detail about various rotations durin insertion of a node in AVL tree.

UNIT IV

17. Discuss the all pairs shortest path algorithm fo finding shortest path in the graph with example.

Or

18. Write algorithm that uses Kruskal's way of finding the minimum cost Spanning tree of a graph.

UNIT V

19. Explain in detail

(a) Hash tables

(b) Inverted tables

Or

20. Write in detail about external storage devices.

UNIT V

19. Compute upto first two series of $f(x)$ given by the

x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$
y	1.98	1.3	1.05

Or

20. Find the complex for $f(x)=e^x$ in $(0,2)$.